Co-Op Support Application

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I. INTRODUCTION

This project involves the development of a small-scale co-op support application designed to assist in managing the co-op admission and reporting process.

The vision of this system is to provide a simple, centralized platform that streamlines interaction between students, coordinators, and employers throughout the co-op lifecycle. The primary goal is to improve efficiency in handling applications, tracking student progress, and collecting required reports and evaluations.

The application supports three main user types: students, employers, and coordinators. Each user is required to create an account using role-specific information:

- Students: Full Name, Student ID, Email, Password
- Employers: Company Name, Supervisor Name, Password
- Coordinators: Email, Password

For Students, the application must provide the following functionality:

- Submit a PDF application
- Check application status (Pending, Accepted, Denied)
- Download work term report template
- Submit a PDF work term report

For Employers, the application must provide the following functionality:

- Submit evaluations for students based on Student ID and in the form of either:
  - PDF
  - Online form with the following fields: Behaviour, Skills, Knowledge, and Attitude

For Coordinators, the application must provide the following functionality:

- View student applications
  - Accept/reject 1st round student applications
  - Send 1st time accepted applicants to a 2nd round of screening
- View student work term reports
  - Send automated reminders to students who have not submitted their reports • View employer reports on their co-op students
- View complete list of applicants, including those accepted or rejected for both provisional and final acceptance

Overall, the system is designed to be simple, functional, and secure, while supporting the core workflow of the co-op process.

II. CUSTOMER REQUIREMENTS (C-REQ):

This section lists the client's requirements, each linked to a system feature.

Req ID	Requirement	Feat. ID	Feature
C-01	Student ID must follow a defined format (e.g. 8 digits) and be validated client-side before submission.	F1	Application Registration
C-02	Each student ID may only be associated with one active application at a time.	F1	Application Registration
C-03	All applicant data must be stored in a persistent database and not lost on server restart.	F1	Application Registration
C-04	Status change timestamps must be recorded.	F1	Application Registration
C-05	Passwords must be a minimum of 8 characters and contain at least one number and one letter.	F1	Application Registration
C-06	Must be different logins for Students, Coordinators, and Employers respectively.	F2	Authentication & Account Management
C-07	Passwords must be hashed before storage; plaintext passwords must never be stored.	F2	Authentication & Account Management
C-08	User must be kicked out of the session after 30 minutes of inactivity.	F2	Authentication & Account Management
C-09	After 5 consecutive failed login attempts, the account must be temporarily locked for 15 minutes.	F2	Authentication & Account Management
C-10	Only provisionally or finally accepted students may create a student portal account.	F2	Authentication & Account Management

C-11	Uploaded report files must be in PDF format only, reject other formats.	F3	Work-Term Report Submission
C-12	Maximum file upload size is 10 MB.	F3	Work-Term Report Submission
C-13	Submission deadline must be enforced; uploads after the deadline must be flagged or blocked.	F3	Work-Term Report Submission
C-14	Each student may only have one active report submission per work term, re-uploads must replace the previous file.	F3	Work-Term Report Submission
C-15	Uploaded PDF's must be stored securely and accessible only to the submitting student and the coordinator.	F3	Work-Term Report Submission
C-16	Employer accounts must require approval by a Coordinator before gaining access to evaluation features.	F2	Authentication & Account Management
C-17	The system must allow Coordinators to send deadline reminder notifications to students.	F3	Work-Term Report Submission
C-18	Employer evaluations must be submitted through the system and stored securely, linked to the appropriate student record.	F4	Employer Evaluation
C-19	Supervisor accounts must be created with a valid email address and verified before they can submit an evaluation	F4	Employer Evaluation
C-20	Evaluation data must be linked to the correct student and work term at the time of submission	F4	Employer Evaluation
C-21	Employers may only submit evaluations for students assigned to their organization	F4	Employer

C-22	Once submitted, evaluations must be read-only and cannot be modified by employers	F4	Employer Evaluation
C-23	A placement rejection must be recorded with a date and an optional reason field	F5	Placement Tracking & Rejection Handling
C-24	The system must distinguish between program rejection (by the coordinator) and placement rejection (by the employer)	F5	Placement Tracking & Rejection Handling
C-25	Each application must have a clearly defined final state (accepted, rejected, withdrawn) and cannot transition after finalization	F5	Placement Tracking & Rejection Handling
C-26	All rejection and acceptance actions must be logged with the acting user's role and timestamp	F5	Placement Tracking & Rejection Handling

III. PRODUCT BACKLOG

This Product Backlog was developed following Scrum principles, with requirements structured as user stories and prioritized based on client needs.

a. Personas

This defines the primary users of the system and their goals to guide the development of relevant features.

Name	Details	Goal
Alex (Student)	A university student applying for co-op and submitting reports	"I want to apply and submit reports"
Mr. Doe (Coordinator)	Reviews applications and manages the co-op process	"I want to manage applications and track submissions"
Ms. Jones (Employer)	Supervises students and submits evaluations	"I want to evaluate students"
CreativeCanvasShop. (n.d.). *Images* [Photos]. Pixabay. https://pixabay.com/users/creativecanvasshop-45016655/		

b. Scenario:

Alex (Student) fills out the co-op application form with their persona details and submits it. After being accepted into the program, Alex logs into the system and uploads a PDF report for his completed work term.

c. Use Case: Submit Co-op Application (Core feature)
The use case describes the system behaviour and interaction flow for a key feature.

Primary Actor	Alex (Student)
Trigger	Alex chooses to apply for co-op
Preconditions	Alex has access to the application form
Main Flow	1. Student opens application form 2. enters name, student ID, and email 3. Student submits the application 4. System validates input 5. System stores application 6. System displays confirmation message
Alternative Flow	• If required fields are missing → system shows error message • If invalid data → system rejects submission
Postconditions	• Success: Application is saved in the system • Failure: Application is not saved

d. User Stories
User stories were written from the perspective of different system users (Students, Coordinators, Employers). The client assigned priorities, while the development team estimated effort in approximate hours. Estimates included a margin of uncertainty to account for variability in implementation and development conditions.

ID	Related Feature / C-Req	As a…	I want to be able to…	So that…	Priority	Effort ± Hours	Status

S1	F1 / C-01, C-02, C-03, C-04	Student	Apply and submit my application for co-op	I can be considered for the program	High	8h ± 2h	Done
S2	F3 / C-11, C-12, C-13, C-14, C-15	Student	Upload my work-term report as a PDF	I can submit my work and is stored securely	High	1h ± 1h	Done
S3	F2 / C-06, C-07, C-08, C-09	Student	Log in with my credentials	I can access my account in the portal	High	7h ± 2h	Done
S4	F2 / C-10	Student	I want to create a portal account after being accepted	I can submit reports	High	5h ± 2h	Done
S5	F3 / C-04	Employer	Submit an evaluation	I can assess the student's performance	High	6h ± 2h	Done
S6	F2 / C-06, C-07, C-08, C-09	Coordinator	Log in securely	I can access coordinator features	High	8h ± 2h	Done
S7	F1 / C-03, C-04	Coordinator	View all applications	I can review applicants	High	6h ± 2h	Done
S8	F1 / C-04	Coordinator	Accept or reject 1st	I can manage	High	5h ± 2h	Done

			round of application s	admissions			
S9	F1 / C-04	Coordinator	Make final acceptance decisions later (2nd round of application s)	Requirements can be rechecked	High	6h ± 2h	Done
S10	F3 / C-13	System	Enforce submission deadlines	Reports are submitted on time	Medium	6h ± 2h	Done
S11	F2 / C-06, C-07, C09, C-16	Employer	Create an account	I can request access and submit evaluations	Medium	6h ± 2h	Done
S12	F3 / C-14, C-15	Coordinator	See which students submitted reports	I can track completion	Medium	6h ± 2h	Done
S13	F3 / C-18	Coordinator	View evaluations for students	They are recorded	Medium	5h ± 1h	Done
S14	F2 / C-16	Coordinator	Approve employer accounts	Only authorized employers can submit student evaluations	Medium	6h ± 2h	Done
S15	F3 / C-13, C-17	Coordinator	Send reminders	Students submit reports on time	Low	6h ± 2h	Done

S16	F2 / C-06, C-07, C-09, C-16	Employer	Create an account	Access employer features	Low	4h ± 1h	Done
S17	F2 / C-05	Student	Have my password validated for complexity (min 8 characters, at least one letter and one number) during registration	My account is secure and meets system requirements	High	1h ± 0.5h	Done
S18	F4 / C-19	Employer	Verify my email address before submitting evaluations	Only verified supervisors can submit evaluations for students	High	5h ± 2h	Done
S19	F4 / C-20	Employer	Have my evaluation automaticall y linked to the correct student and work term when I submit it	Records are accurate and there's no mismatch	High	4h ± 1h	Done
S20	F4 / C-21	System	Restrict employers to only evaluate students assigned to their organization	Employers cannot submit evaluations for students they don't supervise	High	5h ± 2h	Done

S21	F4 / C-22	System	Make submitted evaluations read-only and accessible for viewing for employers	Evaluation integrity is preserved after submission	Medium	3h ± 2h	Done
S22	F5 / C-23	Employer	Record a placement rejection with a date and an optional reason	Rejected placements are documented with context	Medium	4h ± 1h	Done
S23	F5 / C-24	Coordinator	Distinguish between program rejections (by coordinator) and placement rejections (by employer)	The type and source of each rejection is clear in the system	Medium	5h ± 2h	Done
S24	F5 / C-25	System	Lock an application into a final state (accepted, rejected, or withdrawn) with no further transitions	Application states are definitive and cannot be accidentally changed	High	4h ± 1h	Done
S25	F5 / C-26	System	Log all acceptance and rejection actions with	There is a full audit trail of who made each	Medium	5h ± 2h	Done

			the acting user's role and a timestamp	decision and when			

e. Story Card
The story card expands a selected user story into implementation details, including tasks, estimates, and development notes.

***Title*: Submit co-op Application (ID: S1)**

[requirement]
As a student **I want to** apply and submit the co-op application (with personal details (name, student ID, email) **so that** I can be considered for the program

[story]
Application submission: Alex is presented with a form where he inputs his name, student ID, and email address. Once submitted, the system stores the application and makes it available for the co-op coordinator, Mr. Doe, to review.

[tasks]
• Design application form UI
• Validate input fields (e.g. email format, required fields)
• Store application data in databse/list
• Create submission confirmation message

[initial priority]
• High (set by client)

[estimate/risk]
• 8h ± 2h
• Low-medium risk (basic form + validation)

[details]
This is a core feature of the system, as all applicants must submit their information before being reviewed. It will be one of the first features implemented in Sprint 1.

f. Test Plan:
Initial testing conducted during Sprint 1 focused on validating core system functionality, including application submission, basic authentication, and input validation. While many foundational features were successfully implemented, several components, such as file uploads, evaluation submissions, and security-related features remained incomplete or untested at that stage.

In Sprint 2, comprehensive testing was performed across all major system functionalities involving students, employers, and coordinators. Test cases were executed to verify application workflows, including student application submission, login authentication, report uploads with file validation, and employer evaluation submissions. Coordinator functionalities such as reviewing applications, managing acceptance decisions, and tracking submissions were also tested.

Most features performed as expected, with test cases confirming correct system behaviour and data handling. Evaluation submissions were successfully stored and linked to the appropriate student records. PDF-based evaluations were viewable in a read-only format, while evaluations submitted through the online form were not viewable and only indicated submission status. This limitation has been noted as part of the current system behaviour.

Security and validation features, including password requirements, duplicate submission prevention, and file type restrictions, were successfully enforced. Session timeout functionality was successfully verified, and users are automatically logged out after a period of inactivity.

Overall, the system demonstrates a high level of functionality and meets the majority of the specified requirements. Minor limitations remain, but do not significantly impact the core workflow of the application.

g. Test Cases
Test cases were designed to validate the functionality of key features using clear, concise steps and expected results. They are presented in a structured, table-based format for clarity and ease of evaluation.

Test ID	Related Story ID	Test Case	Expected Result	Actual Result	Status

T1	S1	Submit Application with valid data	Application saved in the Database, and its input (name id, email) should be queryable with: SELECT * FROM student_application WHERE student_id= id	Saved and Queryable	Passed
T2	S1	Submit Application with missing fields	An error message is displayed	A unique error message is displayed for the respective missing field	Passed
T3	S7	View list of applications	Applications are displayed	Applications are displayed	Passed
T4	S8	Accept/reject application	Status updated with timestamp in the database	Status, timestamp updated and recorded	Passed
T5	S6	Log in with valid credentials	User logs in successfully	User logs in successfully	Passed
T6	S2	Upload PDF report	File uploaded successfully	File not uploaded	Passed
T7	S11, S16	Create an Employer account	Account is created, shown in the database	Shown in the database	Passed

T8	S13	Submit an evaluation	Evaluation is recoded	N/A	Passed
T9	S6	Login with invalid password	Error message displayed; failed attempt counted	Error message displayed; failed attempt counted	Passed
T10	S6	Login with 5 failed attempts	Account locked for 15 min	N/A	Passed
T11	S6	Session inactivity >30min	User logged out automatically	N/A	Passed
T12	S2	Upload non-PDF report	Upload rejected	Upload rejected	Passed
T13	S2	Upload report > 10MB	Upload reject	N/A	Passed
T14	S2	Upload the report after the deadline	Upload blocked or flagged	Upload flagged on the coordinator side	Passed
T15	S1	Submit a duplicate application	Submission Blocked	Submission Blocked	Passed
T16	S6	Password is less than 8 chars or missing letters/ numbers	Registration/ login rejected	Registration & login rejected, error message displayed	Passed

T17	S8, S9	Accept/reject 1st round applicants	Move accepted applicants to 2nd round	N/A	Passed
T18	S17	Register with invalid input (Password less than 8 characters OR missing a number OR missing a letter)	Registration rejected with error message	N/A	Passed
T19	S17	Register with a valid password (8+ characters, at least one letter and one number)	Account created successfully	N/A	Passed
T20	S18	Employer attempts to submit an evaluation without verifying their email	Submission blocked with prompt to verify email first	N/A	Passed
T21	S18	Employer verifies email via verification link and then submits evaluation	Email marked as verified, evaluation submission permitted	N/A	Passed
T22	S19	Employer submits an evaluation selecting a student and work term	Evaluation record in database is linked to the correct student ID and work term	N/A	Passed

T23	S19	Submit an evaluation for a selected student and work term	Returns the correct student ID and work term matching what was entered (verifying by querying the database): SELECT student_id, work_term FROM evaluations WHERE evaluation_id = id	N/A	Passed
T24	S20	Employer attempts to submit an evaluation for a student not assigned to their organization	Submission blocked with error message	N/A	Passed
T25	S20	Employer submits an evaluation for a student assigned to their organization	Evaluation submitted and saved successfully	N/A	Passed
T26	S21	Employer attempts to edit a previously submitted evaluation	Edit action is unavailable or blocked; submitted evaluation remains unchanged; new submissions are treated as separate entries	N/A	Passed

T27	S21	Coordinator views a submitted evaluation in PDF format	Evaluation is displayed as read-only with all original data intact	N/A	Passed
T28	S21	Coordinator attempts to view a submitted evaluation created through the online form	Evaluation content is not viewable; only submission status is displayed	N/A	Passed
T29	S22	Employer records a placement rejection with a date and a reason	Rejection saved in database with date and reason fields populated	N/A	Passed
T30	S22	Employer records a placement rejection with a date but no reason (optional field left blank)	Rejection saved in database with date populated and reason field null/empty	N/A	Passed
T31	S23	Coordinator rejects a student application (program rejection)	Rejection recorded with type "program" and coordinator's role	N/A	Passed
T32	S23	Employer rejects a student placement (placement rejection)	Rejection recorded with type "placement" and employer's role	N/A	Passed

T33	S23	View a student record that has both program and placement rejections	Displays both rejection types clearly (appears on separate pages), distinguishing between “program” and “placement” along with the acting role (verifying by querying the database): SELECT rejection_type, acting_role FROM rejections WHERE student_id = id	N/A	Passed
T34	S24	Coordinator attempts to change the status of an application already marked as "accepted", “rejected”, or “withdraw” (finalized)	Status change blocked; application remains in final state	N/A	Passed
T35	S25	Coordinator accepts a student application	Audit log entry created with coordinator role, action “accept”, and timestamp	N/A	Passed
T36	S25	Employer records a placement rejection	Audit log entry created with employer role, action "reject," and timestamp	N/A	Passed

T37	S25	View the audit log after multiple acceptance and rejection actions	Displays all logged actions (accept/reject) with correct acting roles and timestamps for the selected student (verified by querying the database): SELECT acting_role, action, timestamp FROM audit_log WHERE student_id=id	N/A	Passed

IV. Sprint 1 Backlog

Backlog contains all combined user stories and their associated tasks, representing the core functionality targeted for the first sprint. This sprint focuses on implementing foundational and high-priority features, while remaining or more complex functionality will be completed in Sprint 2.

User Story	Not Started	In Process	To Verify	Done
S1 **Submit Application**	—	—	—	TASK 1 **Design Application Form**
S15 **Send Reminders**				TASK 2 **Reject and Accept student's application, with timestamp**
S2 **Send Reminders**	—	TASK 3 **Rejecting files that do not have a .PDF extension**	—	TASK 4 **Implement basic UI form**
		TASK 12 **Evaluation PDF file submission**		
S3 **Login Securely (Student)**	TASK 5 **Lockout and session timeout**	TASK 6 **Implementing UI to create a new coordinator**	—	TASK 7 **Database and account authentication/ validation**

S6 **Login Securely (Coordinator)**		**account instead of internally implementing**		TASK 8 **Database successfully storing student student info securely, using hash functions to encode**
S4 **Create Portal Account after Acceptance**	–	–	–	TASK 9 **Student's account is automatically made once accepted**
S5 **Submit Evaluations (Employer)**	TASK 10 **Create a UI to write an evaluation on the website instead of uploading a file**	TASK 11 **Database implementation to store student evaluations**	–	–
S13 **Submit Evaluations (Coordinator)**		TASK 12 **Evaluation PDF file submission**		
S7 **View applications**	TASK 13 **Review and testing**	–	–	TASK 2 **Reject and Accept student's application, with timestamp**
S8 **Accept/ Reject Applications**				
S9 **Accept Applications**				

S10 **Enforce Submission deadlines (System)**	**Not Started**	—	—	—

V. Sprint 2 Backlog

Sprint 2 focused on completing the remaining core and supporting features of the co-op support application, as well as finalizing testing and system validation. This included implementing Employer evaluation workflows, strengthening security features, and ensuring all user roles could interact with the system as intended.

User Story	Not Started	In Process	To Verify	Done
S2 **Upload Work-Term Report**	—	—	—	TASK 1 **Implement file upload validation and storage**
S3 **Login Securely (Student)**	—	—	—	TASK 2 **Complete login and session handling**
S5 **Submit Evaluations (Employer)**	—	—	—	TASK 4 **Implement deadline restriction logic**
S6 **Coordinator Login**	—	—	—	TASK 3 **Finalize coordinator login system**

S10 **Enforce Submission Deadlines**	—	—	—	TASK 15 **Implement deadline restriction logic**
S13 **View Evaluations (Coordinator)**	—	—	—	TASK 12 **Enable viewing of submitted evaluations**
S14 **Approve Employer Accounts**	—	—	—	TASK 6 **Implement employer approval system**
S15 **Send Reminders**	—	—	—	TASK 18 **Implement reminder functionality**
S16 **Employer Account Creation**	—	—	—	TASK 5 **Complete employer registration system**
S17 **Password Validation**	—	—	—	TASK 7 **Enforce password complexity rules**
S18 **Email Verification**	—	—	—	TASK 8 **Implement email verification process**

S19 **Link Evaluation to Student**	—	—	—	TASK 10 **Link evaluations to student and work term**
S20 **Restrict Employer Access**	—	—	—	TASK 9 **Restrict evaluation access to assigned students**
S21 **Read-only Evaluations**	—	—	—	TASK 11 **Prevent editing after submission**
S22 **Placement Rejection**	—	—	—	TASK 13 **Record placement rejection details**
S23 **Distinguish Rejection Types**	—	—	—	TASK 14 **Differentiate rejection types**
S24 **Final State Locking**	—	—	—	TASK 16 **Lock application after final decision**
S25 **Audit Logging**	—	—	—	TASK 17 **Log system actions with timestamps**